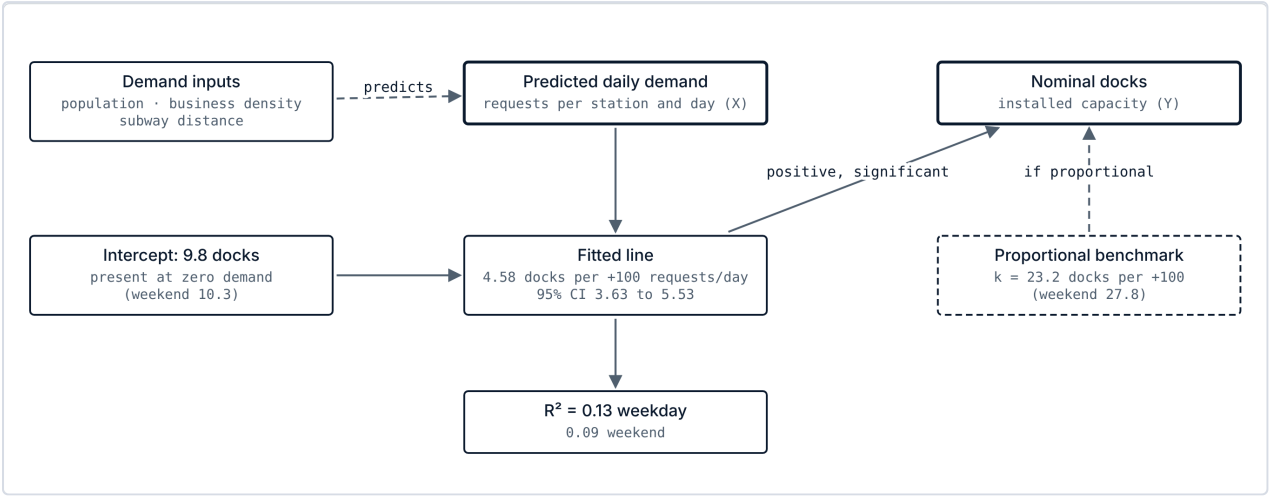
The system is bigger than the model.

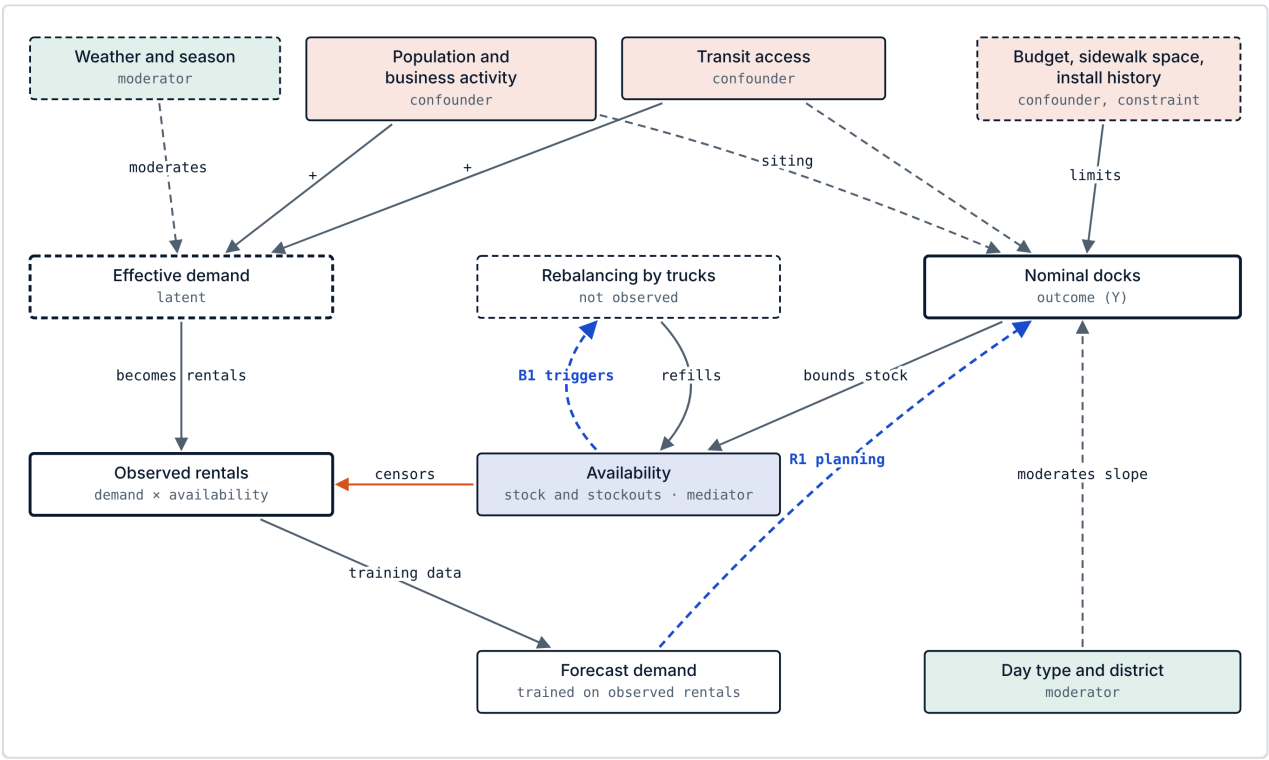
Map A shows what the regression supports. Map B adds theory and hidden features. Map C shows how predictions become an allocation. Arrows show evidence status, not proven causation.

MAP A · REGRESSION EVIDENCE



Nominal docks were regressed on predicted daily demand per 100 requests, with HC3 robust errors. Weekday slope 4.58 docks (95% CI 3.63 to 5.53), intercept 9.79 (9.32 to 10.25), R^2 0.13. The proportional benchmark has slope k = 23.2 and intercept 0. Weekend: 4.27 (3.48 to 5.05), intercept 10.32, k = 27.8, R^2 0.09. The relation is positive and significant, and far from proportional. **Verdict on H_0.** The HC3 Wald test of the zero-intercept restriction gives $\chi^2(1)$ = 1,731.7 (weekday) and 3,463.3 (weekend), $p < 0.001$, so strict proportionality is rejected at the station level. The joint test with k held fixed gives $\chi^2(2)$ = 1,732.7 and 3,810.6 and is treated as exploratory because k is estimated from the same data. Predicted demand is a proxy, so this describes association, not cause. **By district** the picture is milder: across 25 districts the dock-share gap to demand share is 9.9% (95% CI 8.9 to 11.4) on weekdays and 9.4% (8.3 to 11.3) on weekends, with a largest single-district gap of 2.7 points (Gangseo, under-supplied relative to demand).

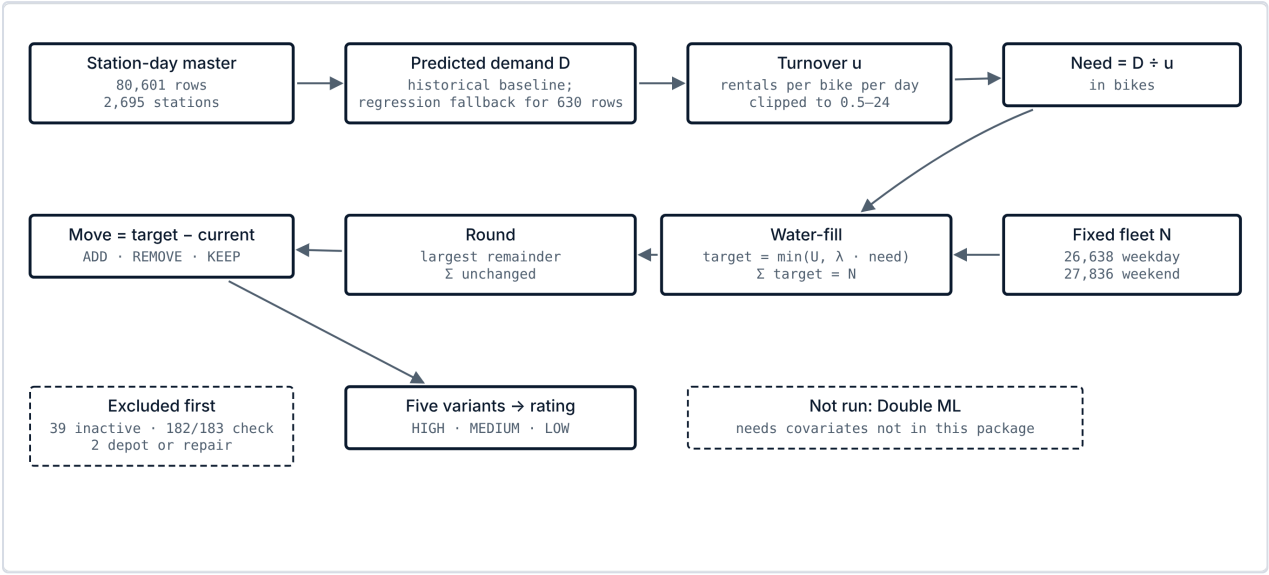
MAP B · THEORY AND HIDDEN FEATURES



Roles. Population and transit access are confounders: they raise demand and also decide where docks are sited. Budget and sidewalk space constrain docks. Availability is a mediator: demand reaches the rental record only through bikes being present, so stockouts hide lost demand. Weather and day type are moderators that change how strongly demand turns into rentals or docks.

Feedback loops. R1 (reinforcing): docks bound stock, stock limits observed rentals, rentals train the forecast, and the forecast guides planning for docks. Under-supplied stations keep looking low-demand. **B1 (balancing):** stockouts trigger truck rebalancing, which refills stock and relieves the stockout. Because trucks already move bikes, midnight stock is not natural supply. Dashed boxes are unobserved or assumed; only the solid boxes carry measured data in this package.

MAP C · PREDICTION AND ALLOCATION



Station-day demand D comes from each station's own normal-day average, with a regression fallback for 630 rows. Turnover u (rentals per bike per day) is clipped to 0.5–24. Need = D ÷ u. A water-filling step sets target = min(upper bound, λ · need) so that targets sum to the existing fleet N (26,638 weekday, 27,836 weekend), then largest-remainder rounding keeps whole bikes. Move = target − current. Five sensitivity variants and data flags give each station a rating. No machine-learning model was trained in this package: Double ML needs covariates that are not included, so Map C describes the allocation pipeline, not a learned model.

WHY A VARIABLE THAT LOOKS WEAK MAY STILL MATTER

- Measurement error. Predicted demand is an estimate and docks are a recorded count. Noise in a predictor pulls its slope toward zero.
- Overlap with other variables. District, transit access and business density move together, so each looks weak when they share one model.
- A straight line averages very different stations, from 0.5 to 708 predicted requests a day. A hub effect can vanish in the average.
- Indirect effects. An effect that works through another variable, such as availability, disappears from the direct coefficient.
- A short window. One month of station averages hides weather and event swings.
- Low explanatory power. With R^2 of 0.13, most of the variation in docks lies outside demand.

A variable can show a small or non-significant coefficient and still change the system. The hidden features below are the ones we cannot test with this data.

SEVEN HIDDEN FEATURES

FEATURE	ROLE	WHY IT MAY MATTER	IF WE IGNORE IT	NEXT TEST
Budget and sidewalk space	Confounder, constraint	Docks sit where installation is easy and affordable, not only where people want bikes.	Targets may ask for bikes where docks cannot be installed. 1,889 of 4,942 targets already exceed nominal docks.	Collect installation limits and treat the dock limit as a hard bound.
Weather and season	Moderator	Demand and stockouts swing from day to day; a monthly mean hides it.	Static targets leave stations empty on dry days and overfull on wet days.	Merge daily weather and test its interaction with day type.
Rebalancing records	Unobserved mediator	Midnight stock already reflects truck moves, so it is not natural supply.	Recommendations may repeat or undo moves crews already make.	Obtain transfer logs and separate organic flow from operator moves.
District and transit overlap	Confounder	Docks, demand and district move together, so each p-value looks weak.	Dropping a weak-looking variable can bias the remaining slope and misdirect additions.	Compare nested models and report variance inflation and effect sizes.

Availability and stockouts	Mediator	Rentals are demand filtered by available bikes.	Demand is under-counted where stations run empty. Stations that gain bikes had a 22% mean stockout rate.	Validate the lost-demand correction before using it for sizing.
Day type and district	Moderator	The relation may differ by day type or district.	One pooled rule can over-serve some districts and under-serve others.	Fit a pooled model with interactions, then test them.
Midnight-only inventory	Measurement	A single daily snapshot misses refills during the day.	Hubs look undersupplied or oversupplied. One station shows 708 requests a day with about one bike at midnight.	Add intraday inventory for the busiest stations.

POLICY IMPLICATIONS IF HIDDEN FEATURES ARE IGNORED

HIDDEN FEATURE	IF IGNORED	EVIDENCE IN THIS DATA	WHAT TO DO
Budget and sidewalk space	The plan asks for bikes where docks cannot be installed.	1,889 of 4,942 targets exceed nominal docks.	Obtain installation limits; keep the dock limit as a hard bound.
Weather and season	Static weekday and weekend targets fail on wet or extreme days.	One month, one season.	Merge daily weather; rerun for other months.
Rebalancing records	Recommendations repeat what crews already do.	Midnight stock reflects past truck moves; daily rebalancing is not observed.	Obtain transfer logs.
Stock and dock mismatch	Wrong dock counts distort every target near them.	1,313 rows flagged for stock above docks; 365 station-weeks left out at 3× or more.	Verify dock counts and the inventory field before acting.
Midnight-only inventory	Hubs are mis-sized.	One station: 708 requests a day, about one bike at midnight.	Add intraday inventory.
Lost demand	Empty stations stay empty because their demand is invisible.	Stations that gain bikes: 22.2% mean stockout rate; stations that give bikes: 2.4%.	Validate the correction before sizing with it.

EQUITY CHECK ACROSS DISTRICTS

Net moves are unrelated to demand per station (rank correlation 0.05 weekday, −0.06 weekend), so low-demand districts are not systematically penalised. Some districts still give up a large share of today's bikes. The baseline is midnight stock, which can overstate gains in central districts, so these figures are a flag for review, not a verdict. Population-weighted equity was not computed here.

DISTRICT	WEEKDAY NET MOVE, % OF CURRENT	WEEKEND NET MOVE, % OF CURRENT
Geumcheon	−18.1	−19.9
Dongdaemun	−16.0	−14.8
Guro	−15.5	−15.0
Jungnang	−14.5	−11.0
Jung-gu	+50.2	+36.2
Jongno	+32.5	+38.1
Gangbuk	+30.8	+25.0
Gwangjin	+24.7	+24.2

Suggested safeguard: test a scenario with a minimum service level per district before any move is made.

Overall: treat the plan as a prioritised list for review, not an order to move bikes. HIGH-rated stations are the safest starting set; LOW-rated stations need data checks first.